

Units of area



1 Consider the following units of measure for area:

- Hectares
- Square centimetres
- Square metres
- Square kilometres

Choose the most appropriate unit of measure for the following areas:

- | | |
|---------------------|-----------------|
| a Postage stamp | b Bathroom wall |
| c Australia | d Matchbox |
| e Exercise book | f Street |
| g School playground | h Pencil case |
| i Japan | j Cabinet |
| k City | l Television |
| m Kitchen | n Farm |
| o Classroom | p Netball court |
| q Table | r National park |
| s Door | |

2 Convert the following to the unit indicated:

- | | |
|---|---|
| a 3 m^2 to square centimetres | b 7 m^2 to square centimetres |
| c 0.56 km^2 to square metres | d 810 mm^2 to square centimetres |
| e $34\,000 \text{ mm}^2$ to square metres | f $790\,000\,000 \text{ cm}^2$ to square kilometres |
| g 6 ha to square metres | h 10.4 ha to square metres |
| i $20\,000 \text{ m}^2$ to hectares | j $54\,800 \text{ m}^2$ to hectares |

3 Find the area of a square of side 290 cm , in square metres.

Applications

4 A park has an area measuring 19 acres. Given that 1 acre is approximately 4000 m^2 , find the area of the park in square metres.

- 5 Rectangular farms around Australia were measured and their dimensions, in metres, are recorded in the following table:



- a Find the area of each farm in square metres.
- b Which farms have an area of more than 1 ha?
- c Which farms have an area of less than 1 ha?
- d Which farm has an area of exactly 1 ha?

Farm	Length, m	Width, m	Area, m ²
1	200	100	
2	350	21	
3	100	24	
4	400	40	
5	100	100	

- 6 A property covers an area of 4.5 ha. Given that 1 ha is 10 000 m², calculate the following:
- a The area of the property in square metres.
- b The area of the property in acres, given that 1 acre is approximately 4000 m².
- 7 A large cattle station covers an area of 320 acres. Given that 1 acre is approximately 0.4 ha, and 1 ha is 10 000 m², calculate the following:
- a The area of the cattle station in hectares.
- b The area of the cattle station in square kilometres.
- 8 A farm covers an area of 51 000 m². Given that 1 ha is 10 000 m², and 1 acre is approximately 4000 m², calculate the following:
- a The area of the farm in hectares.
- b The area of the farm in acres. Round your answer to two decimal places.
- 9 Bob owns a paddock with an area of 290 acres. He wants to advertise his eggs as free range, with no more than 400 chickens per hectare. Given that 1 acre is approximately 0.4 ha, find the largest amount of chickens that Bob can keep in his paddock.
- 10 A farm covers an area measuring 340 acres. Given that 1 acre is approximately 0.4 ha, and 1 ha is 10 000 m², calculate the following:
- a The area of the farm in hectares.
- b The area of the farm in square metres.
- 11 American football is played on a field with an area of 12 000 yd².
- a Given that 1 yd² is approximately 0.836 m², find the area of an American football field in square metres.

- b** The MCG oval has an area of $17\,700\text{ m}^2$. How much smaller than the MCG oval is an American football field? Give your answer in square metres.



- 12** Manhattan, the most densely populated borough of New York City, has an area of 22.8 mi^2 . If 1 mi^2 is approximately 2.6 km^2 , how large is Manhattan in square kilometres?
- 13** The state of California is roughly $164\,000\text{ mi}^2$ in area. Given that 1 mi^2 is approximately 2.6 km^2 :
- a** Find the area of California in square kilometres.
- b** The state of New South Wales has an area of approximately $809\,000\text{ km}^2$. How much larger is New South Wales than California?
- 14** An official NBA court covers 4700 ft^2 in area. Given that 1 m^2 is approximately 10.8 ft^2 , find the area of an official basketball court in square metres. Round your answer to three decimal places.